Module 2: Relationships Between Variables — Practice

Lesson 4 | State the denominator and interpret the pattern

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Preparation

Read Lesson 4. Use Restaurant and Electronics from the checked CSV bundle. Work in the `20261` project and submit a short Quarto report with reproducible code and interpretations.

```
# read.csv() imports a CSV into a data frame with named columns.
# The relative path starts at the course project folder; the source CSV is not modified.
restaurants <- read.csv("data/module02/restaurant_ratings.csv")
electronics <- read.csv("data/module02/electronics_sales.csv")
```

1. Construct and check the crosstab

1. Identify the observational unit and classify the two Restaurant variables.
2. Group prices into [10,20), [20,30), [30,40), [40,50) USD. Explain where a price of exactly USD 30 belongs.
3. Construct the quality-by-price-band crosstab, retaining the order Good, Very Good, Excellent. Include marginal totals and verify that the grand total is 300.
4. Interpret the cell containing 46 and its row and column totals.
5. Explain what information price grouping loses compared with the original numerical values.

2. Choose the denominator

Calculate each percentage from counts. For every answer, state the numerator, denominator, and reference group.

1. All restaurants: Very Good **and** priced USD 30 up to, but below, USD 40.
2. Among Very Good restaurants: the proportion in that band.
3. Among restaurants in that band: the proportion rated Very Good.

4. Among Excellent restaurants: the proportion priced at least USD 30. Compare it with the corresponding proportion among Good restaurants.
5. Among restaurants in the highest band: the proportion rated Excellent. Explain why this differs from the proportion in the highest band among Excellent restaurants.
6. Draw percentage bars comparing price distributions within quality categories. Show group sizes and identify which dimension totals 100%.

3. Inspect paired numerical observations

1. Identify the observational unit and explain the sales units in Electronics. Convert the first week's sales to USD.
2. Draw a scatterplot of commercials against sales. Describe direction, form, strength, and any observations that might deserve investigation.
3. Explain why sorting the two columns independently invalidates the analysis.
4. Explain why putting commercials on the horizontal axis does not establish causation.

4. Calculate and interpret association

1. Calculate both sample means and the sum of cross-products of deviations. Use them to calculate sample covariance and check with `cov()`.
2. State covariance's units and explain its sign.
3. Calculate Pearson correlation using covariance and the two sample SDs. Check with `cor()` and interpret it in context.
4. Convert sales to USD. Recalculate covariance and correlation and explain the difference in behavior.
5. Evaluate: "A correlation of 0.93 means each extra commercial raises sales by 93%."
6. Create `u <- -4:4` and `v <- u^2`. Draw the scatterplot and calculate correlation. Explain what the example shows about zero correlation.

5. Write a responsible conclusion

In three or four sentences, summarize the Electronics result, give one plausible alternative explanation to investigate, and explain what the ten observations do not establish. Label the alternative explanation as a possibility, rather than a fact in the file. Keep this conclusion descriptive; no hypothesis test or regression model is required.

Return to Lesson 4. Then complete the integrated review.